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## Mapping apparatus and method

### Abstract


The instant application provides a system comprising at least one or more receiver units and one or more targets which work in conjunction with a users' existing aerial drone to improve the accuracy of photogrammetric data collection.



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## Field of Classification Search

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of, and priority to, U.S. Provisional Patent Application No. 63/155,775, filed on Mar. 3, 2021, titled “Mapping Apparatus and Method.”

FIELD OF THE DISCLOSURE
(1) The present invention is directed to a system and method for obtaining accurate topographic surveying position data, such as photogrammetric data, quickly and in a cost-effective manner. Further, the present invention allows the quick and easy use of the data obtained by a receiver during a survey.
BACKGROUND
(2) Aerial mapping using drones is well known in the art. However, the current solutions do not provide cost-efficient technologies that produce easy to use and manipulate data files. For example, some of the Global Navigation Satellite System (GNSS)/Global Positioning System (GPS) receivers on the market today cost upwards of $20,000 each, and multiple receivers may be needed to complete one mapping project. The present system detailed herein solves the issues associated with the prior art solutions.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) To further illustrate the advantages and features of the present disclosure, a more particular description of the invention will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. It is appreciated that these drawings are not to be considered limiting in scope. The invention will be described and explained with additional

specificity and detail through the use of the accompanying drawings in which:
(2) FIG. **1** shows one embodiment of a receiver unit described in the instant application.
(3) FIGS. **2**A and **2**B show various alternate embodiments of the targets described in the instant application.
(4) FIG. **3** shows one embodiment of placing a receiver unit on a target as described in the instant application.

DETAILED DESCRIPTION

(5) As shown in FIGS. **1-3**, the present system **10** provides an apparatus and method of use to obtain accurate and easily usable survey data. The system **10** includes one or more receiver units **20** that are used as ground control points to work in conjunction with a users' existing aerial drone **110** to improve the accuracy of photogrammetric data collection. These receiver units **20** collect data from the GNSS/GPS system and use that data with correction data using established National Geodetic Service (NGS) Continuously Operating Reference System (CORS) data to post-process kinematic (PPK) solutions for improved accuracy using desktop powered software solution.
(6) The receiver unit **20** is especially useful in providing surveying and mapping services and with aerial drones **110** during aerial photogrammetry. In one embodiment, the receiver unit **20** comprises a positioning antenna **30** attached to a ferrous metal grounding plate **25** which prevents multi-phase interference **40**. In one embodiment, the positioning antenna **30** is a ZED-F9P multi-band GNSS/GPS manufactured by u-Blox Holding AG, although other positioning antennas **30** that are capable of similar functions should be considered within the scope of this disclosure. The receiver unit **20** is built of a single board designed to also include several other useful features such as a charging control module **50** to control the discharge of a battery **60**, a secure digital (SD, or micro-SD) card reader **70**, a battery charging port **80**, and an on/off switch or button **90**.
(7) The system **10** also includes one or more ground targets **100** that are designed to be visible to the aerial drone **110** from the air. The targets **100** may optionally include a contrasting color scheme to make it easier to locate the target **100** from the air via the aerial drone's **110** camera **120** which is taking photographs of the property. In one embodiment, the target **100** comprises a first color **130** and a second color **140**, which may be contrasting. In one embodiment, one side of the target **100** includes a different contrasting color scheme than the opposite side. The target **100** is manufactured of flexible materials to allow easy field conveyance, setup, and retrieval.
(8) One exemplary use of the system **10** will be described below.
(9) Initially, a user **150** would place multiple targets **100** about a property **160** to be aerial surveyed or mapped. Then a receiver unit **20** would be placed in the center of each target **100**. The receiver units **20** collect position data via the positioning antenna **30** and store said data. It is known in the art the positioning data gathered by the receiver unit **20** may need to be corrected via some method. Then, once the areal drone **110** has completed its course which covers the property **160**, the receiver units **20** and the targets **110** are retrieved by the user **150**. When the aerial drone **110** flight is complete, the data stored on the receiver units **20** is downloaded and analyzed.
(10) After collecting the targets **110** and the receivers **20**, the data contained on the receivers **20** is downloaded to a computer **170** that is running a software solution application **180** designed to interact with the receiver units **20** and provide, in part, PPK processing, including determining where the receiver units **20** were positioned. Using the software solution application **180**, the nearest CORS is found and the software solution application **180** connects with the CORS servers to retrieve the overlapping corrections files therefrom. The PPK processing corrects any errors generated from the GPS data collected in the field by the receiver units **20**. The software solution application **180** of the present invention provides many advantages over the prior art solutions. Initially, the software solution application **180** allows desktop-based map window view of the project site address, field conditions and mission planning to estimate flight time and required overlaps. After flight, the software solution application **180** collects and manages the data files from the receiver units **20**, organizes the files into a project, identifies up to five (5) available

CORS files, and collates the data files that can be easily overlayed onto a photo mapping solution such as those known in the art. The present disclosure provides several advantages over the prior art, including: Dramatically speeding the processing of photogrammetric data collection and reducing the overall costs of the collection process. Providing a simple and rugged receiver unit **20**. Improving visibility of the targets **100** for airborne cameras based on varying terrain conditions. For instance, on concrete or asphalt surfaces, the orange/black checker is optimum where white/black checker would fade as the camera is at higher altitudes. Conversely, on green grass and similar surfaces, the white/black grid has great contrast to the surrounding environment and is easily registered on the camera, where orange/black colors will fuse and in reflective lighting, disappear in the image.
(11) The use of “adapted to” or “configured to” herein is meant as open and inclusive language that does not foreclose devices adapted to or configured to perform additional tasks or steps. Additionally, the use of “based on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based on” one or more recited conditions or values may, in practice, be based on additional conditions or value beyond those recited. Headings, lists, and numbering included herein are for ease of explanation only and are not meant to be limiting.
(12) The term “property” as used herein should not be construed to limit the term to a legally defined area or parcel, but rather more generally to an area of interest to be topographically mapped.
(13) The terms “about” and “approximately” shall generally mean an acceptable degree of error or variation for the quantity measured given the nature or precision of the measurements. Typical, exemplary degrees of error or variation are within 20 percent (%), preferably within 10%, more preferably within 5%, and still more preferably within 1% of a given value or range of values. Numerical quantities given in this description are approximate unless stated otherwise, meaning that the term “about” or “approximately” can be inferred when not expressly stated. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.
(14) Although particular embodiments of the present disclosure have been described, it is not intended that such references be construed as limitations upon the scope of this disclosure except as set forth in the claims.

## Claims

1. A method of conducting a topographic survey of a property, said method comprising: (i) placing one or more flexible targets above stationary ground about the property, wherein said target comprises a contrasting color pattern comprised of at least two (2) contrasting colors in a checker board pattern; (ii) placing a receiver unit on top of said above ground targets, wherein said receiver unit comprises at least a positioning antenna, a ferrous metal grounding plate and a non-volatile memory to store position data received by the positioning antenna; (iii) operating an aerial drone which takes photographs of the property; (iv) retrieving data from a stationary continuously operating reference station with a known and fixed location (v) performing post-kinematic processing on the stored position data using at least data gathering from the continuously operating reference station and the photographs taken by the aerial drone which determines the position of the one or more flexible targets about the property.
2. The method of claim 1 wherein the post-kinematic processing determines where the receiver unit was positioned.
3. The method of claim 1 wherein the drone flies a grid pattern over the property.
4. The method of claim 1 wherein data is retrieved from more than one continuously operating reference station.
5. The method of claim 1 wherein at least four (4) targets and receivers are used to survey the

property.
6. The method of claim 1 wherein the data is retrieved from the continuously operating reference station to the receiver.
7. The method of claim 1 wherein the contrasting color pattern includes orange and black colors.
8. The method of claim 1 further comprising retrieving data from at least three (3) stationary continuously operating reference stations with known and fixed locations.
9. The method of claim 8 further comprising retrieving data from five (5) stationary continuously operating reference stations with known and fixed locations.